Supplementary Step 8 Visualization Report

Overview

Metric	Value
Tested concrete-abstract pairs	225
Pairs surviving FDR	0
Matched-pair subset size	12
Matched subset FDR-significant pairs	0
Ranked questions	30
Selected channel-followup pairs	3

Interpretation note. The exploratory item-pair analysis did not identify any concrete-abstract question pairs with reliable back-ROI differences after Benjamini-Hochberg correction. This supports the view that the back ROI does not show robust item-specific discrimination in the current dataset. The lowest-q pair was 2018_CH_77 versus 2021_CH_86 (mean difference=1.864e-07, raw p=0.003431, q=0.596248). No matched-pair subset comparison survived FDR correction under the predeclared matching rule.

Top pair summary. Top descriptive pair: 2018_CH_77 versus 2021_CH_86 (mean difference=1.864e-07, raw p=0.003431, q=0.596248).

Core Step 8 Heatmaps

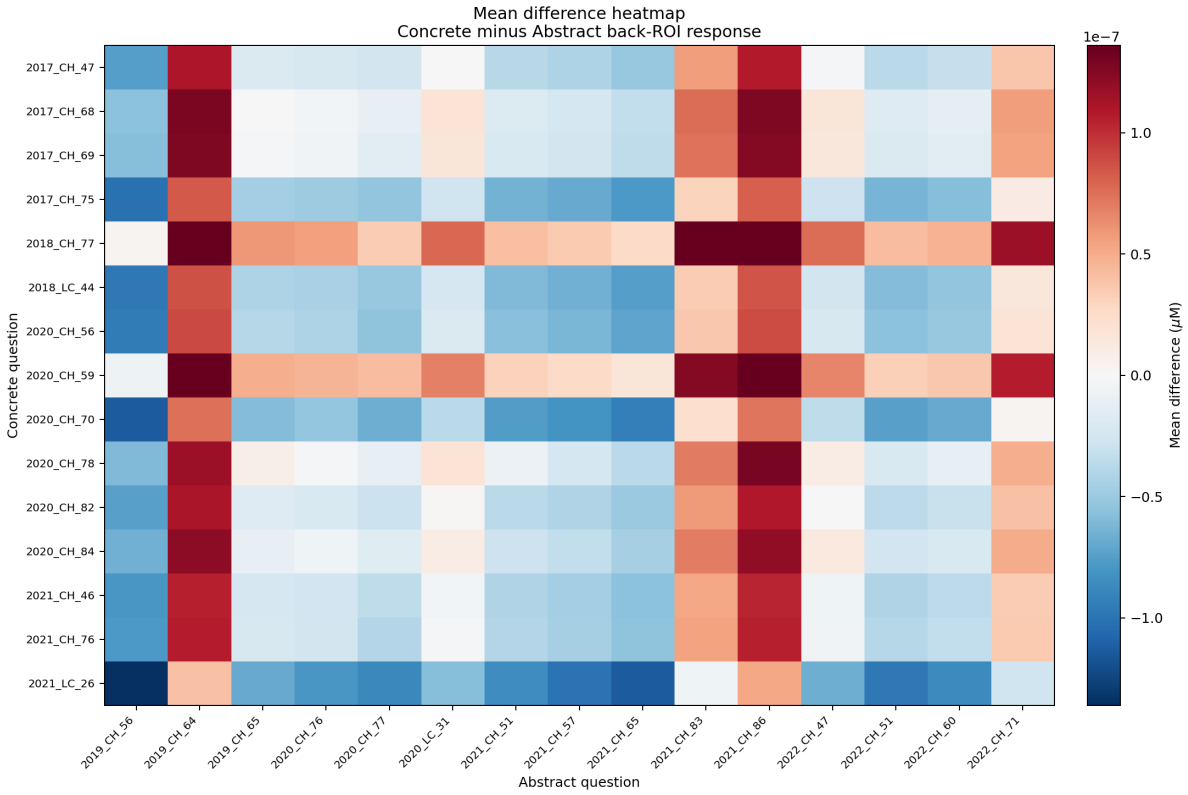

Figure 1: Mean back-ROI concrete-minus-abstract differences for all item pairs.

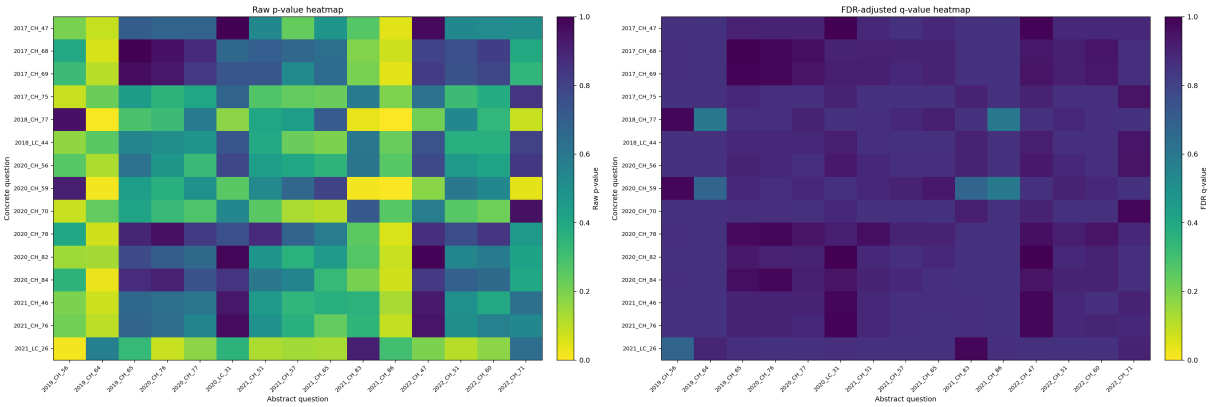

Figure 2: Raw p-values and FDR-adjusted q-values across all tested item pairs.

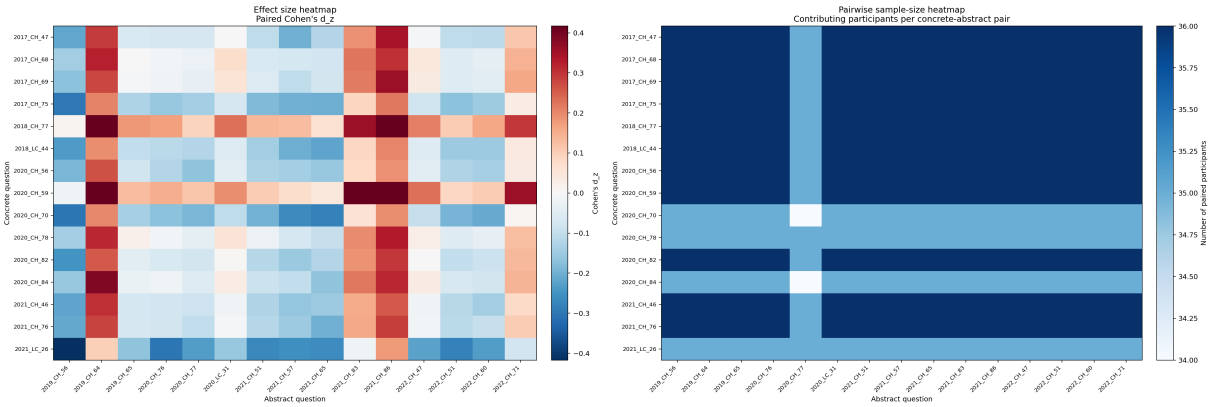

Figure 3: Paired effect sizes and the contributing sample size for each concrete-abstract pair.

Descriptive Ranking Views

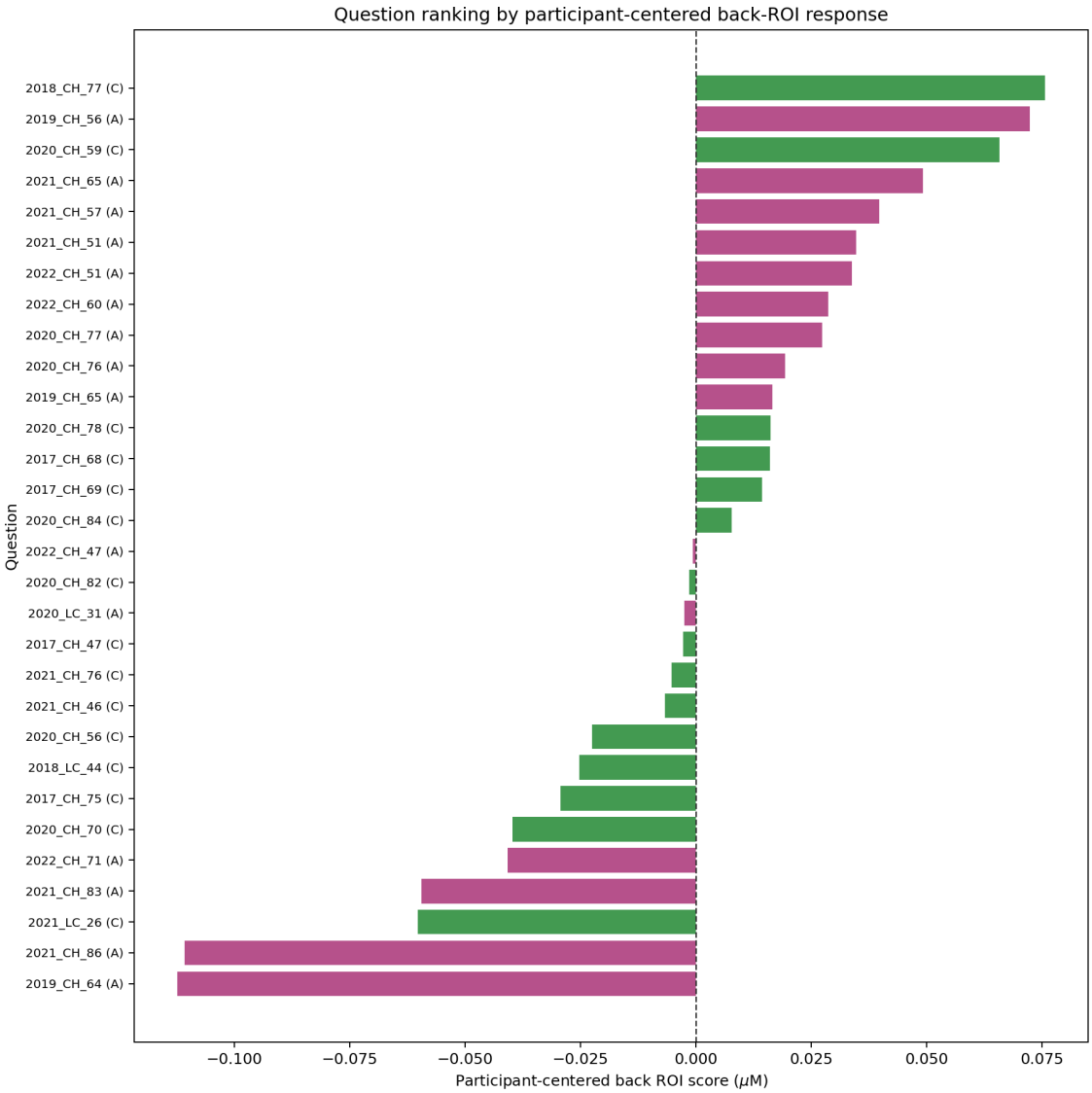

Figure 4: Participant-centered ranking of questions by back-ROI response.

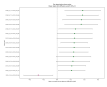

Figure 5: Top descriptive item pairs ranked by the strongest pairwise evidence, shown with mean differences and 95% confidence intervals.

Matching And Channel Follow-up

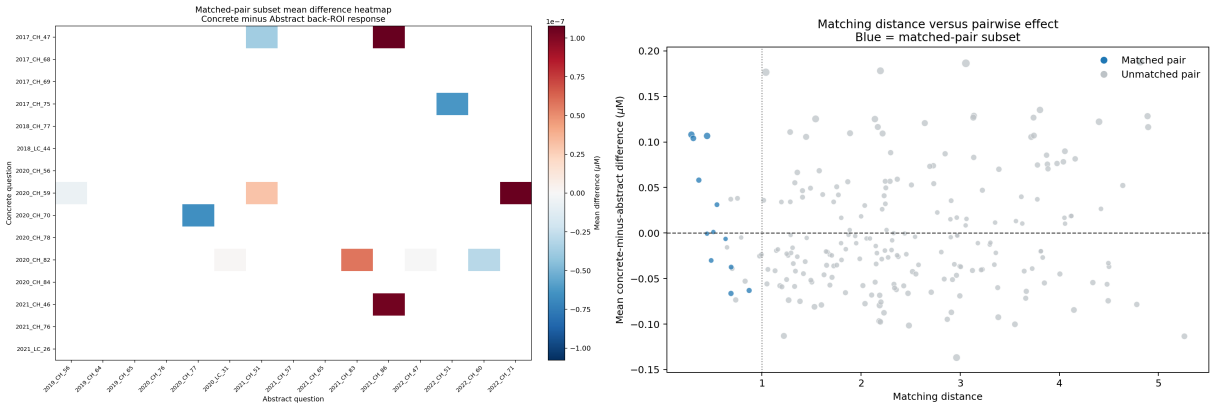

Figure 6: Matched-pair subset heatmap and the relation between matching distance and pairwise mean effect.

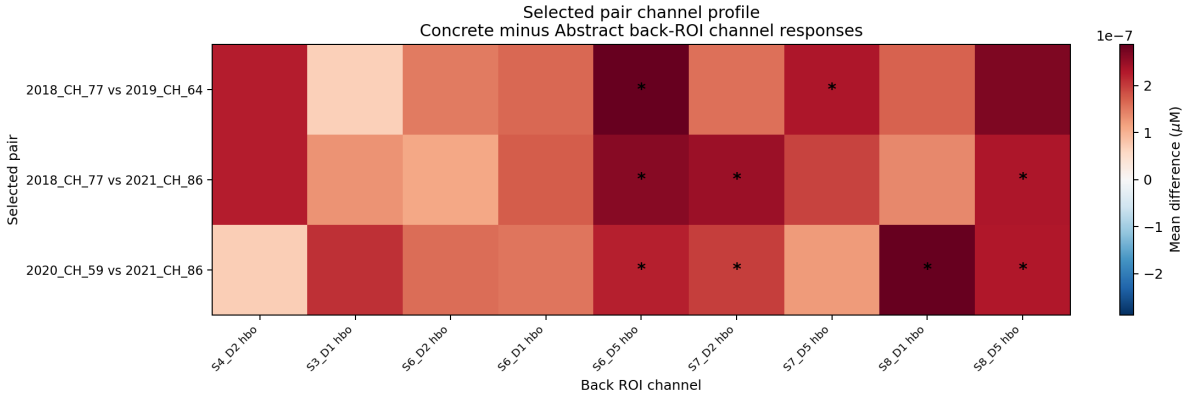

Figure 7: Optional back-channel follow-up for the selected descriptive pairs.